

Retail Performance Intelligence Hub

DAX Measures Documentation

Self-Initiated Portfolio Project | Power BI Business Analytics

1. Purpose

This document records the DAX measures used in the Retail Performance Intelligence Hub Power BI dashboard. The measures support KPI cards, trend analysis, profitability analysis, product analysis, regional analysis and customer-segment analysis.

The project uses a synthesized retail dataset for portfolio demonstration purposes. Keep these formulas aligned with the Power BI model if the report is modified later.

2. Core DAX Measures

1. Total Revenue

```
SUM(Orders[Sales])
```

Calculates total sales/revenue in the current filter context.

2. Total Cost

```
SUM(Orders[Cost])
```

Calculates total cost in the current filter context.

3. Total Profit

```
SUM(Orders[Profit])
```

Calculates total profit in the current filter context.

4. Profit Margin

```
DIVIDE([Total Profit], [Total Revenue], 0)
```

Calculates profit as a proportion of revenue; returns 0 when revenue is zero.

5. Order Count

```
DISTINCTCOUNT(Orders[Order_ID])
```

Counts unique orders in the current filter context.

6. Average Order Value

```
DIVIDE([Total Revenue], [Order Count], 0)
```

Calculates average revenue per order.

7. Return Rate

```
DIVIDE(CALCULATE([Order Count], Orders[Returned] = "Yes"), [Order Count], 0)
```

Calculates the proportion of orders marked as returned.

8. On-Time Shipping %

```
DIVIDE(CALCULATE([Order Count], Orders[On_Time] = "Yes"), [Order Count], 0)
```

Calculates the proportion of orders marked as shipped on time.

9. Average Shipping Days

```
AVERAGE(Orders[Shipping_Days])
```

Calculates the average shipping duration in days.

3. Measure Usage Across the Dashboard

Measure	Primary analytical use
Total Revenue	Revenue KPIs, trends, region/category/product/segment comparisons
Total Cost	Cost analysis and profitability context
Total Profit	Profit KPIs, trends, product/category/region/segment comparisons
Profit Margin	Profitability KPI and regional/product/category profitability analysis
Order Count	Order KPI, order trends and order-volume comparisons
Average Order Value	Order-value analysis and regional/customer-segment comparisons
Return Rate	Operational KPI and return monitoring
On-Time Shipping %	Operational performance monitoring
Average Shipping Days	Shipping-speed monitoring

4. Filter Context

The measures respond to the dashboard's interactive filter context. When users filter by dimensions such as Year, Region, Category, Segment or Product_Name, the measures recalculate for the selected context.

This makes the KPI cards and visualizations interactive rather than hard-coded to a single overall value.

5. Formatting Guidance

Currency: Total Revenue, Total Cost, Total Profit and Average Order Value should use the same currency convention throughout the report.

Percentage: Profit Margin, Return Rate and On-Time Shipping % should be displayed as percentages.

Counts: Order Count should be displayed as a whole-number count.

6. Portfolio Documentation Note

These measures are documented as part of a self-initiated Power BI portfolio project. They demonstrate practical use of DAX for KPI creation, business performance analysis and interactive dashboard reporting.

Project disclaimer: This project uses a synthesized dataset created for portfolio and demonstration purposes. It does not represent client work, confidential business information, or actual company performance.